

Computational Physics - Projects

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A note about the projects

Projects and guidance uploaded to Blackboard Monday at 12pm.

- Word and LaTeX templates for the project report
- Three different projects:
 - Project 1: A log-likelihood fit for extracting neutrino oscillation parameters
 - Project 2: The building blocks of planet formation
 - Project 3: Quantum Monte Carlo

A note about the projects

- Number of tasks in each project varies – **the amount of work does not!**
 - Some projects describe the work in detail
 - Project 1, for example

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- Number of tasks in each project varies – **the amount of work does not!**
 - Some projects describe the work in detail
 - Project 1, for example
 - Other projects provide more freedom for you to determine the work
 - Project 2, for example
- Please discuss your project with the demonstrators if you are unsure

A note about Project 1

- Project 1 uses bespoke data files for each student, generated from the real experimental Monte Carlo simulation
 - The true neutrino oscillation parameter values are varied
 - Independent statistical fluctuations are applied
- As such, you will (should) not get the same answer as anyone else on the course!
- If you cannot access your data file please let me (or a demonstrator) know
 - Students at IC last year should have a file
 - Students that started at IC this year will not – let Mark Scott know if you want to do Project 1 and I will create one

Project Choice

- We have a survey on Blackboard to record your project choice
- Please fill this in once you have chosen a project. If you change your mind it doesn't matter. You don't need to let us know.

Project Submission

- Project reports and code are submitted to Blackboard
 - Process described in the guidance notes
 - Both report and code are checked for plagiarism
- **Deadline for submission: Friday December 12th at 5pm**
- Submission areas will be made available well in advance of the submission deadline
 - Recommend you submit with plenty of time remaining to avoid technical issues

ChatGPT and Plagiarism

- Departmental Policy on Plagiarism
 - Submission of code, text, figures (and ideas!) without acknowledging they were created by someone else
 - If submitted accidentally it's still plagiarism, and treated as such
 - Treated very seriously by College
- Generative AI tools (Github CoPilot, ChatGPT etc.)
 - Follow the department's guidelines, which are based on the College's guidance.
 - Use of AI without acknowledgment is plagiarism
 - AI is pretty good at coding
 - AI is pretty bad at validation, testing and understanding physics results
 - Your grade is based on the second one, not the first...

Project Questions

- The project [guidance](#) provides detailed instructions on the project submissions and marking
- If you have any questions, please ask during the practical sessions or on the [Ed Discussion](#) board
- Using in-built methods is allowed **provided the method does not give the solution to a task**
 - If we ask you to do something in the script you should write the code to do this
 - If you need to use another method to complete a task (e.g. matrix multiplication) then it is fine to use a built-in function
 - If in doubt ask!